

TMREF-AD: Trust-Aware Multimodal Regional Expert Fusion for Alzheimer’s Disease Diagnosis

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Abstract—Alzheimer’s disease (AD) classification from multimodal neuroimaging remains challenging due to regional disease heterogeneity and the varying reliability of imaging biomarkers across disease stages. Existing MRI-PET approaches commonly rely on global feature aggregation or simple feature concatenation, which often overlook region-specific pathological patterns and modality-specific contributions. To address these limitations, we propose TMREF-AD, a trust-aware regional mixture-of-experts framework for multimodal Alzheimer’s disease classification. TMREF-AD first partitions the brain into five clinically meaningful anatomical regions using FastSurfer-based parcellation. Regional MRI and PET representations are extracted through modality-specific 3D convolutional neural networks and refined using intra-modal attention to capture inter-regional dependencies within each modality. The enriched regional representations are subsequently fused and processed by specialized regional experts, whose predictions are aggregated through a trust-aware fusion mechanism that integrates routing weights, prediction confidence and expert reliability. Experimental evaluation on the ADNI dataset demonstrates that TMREF-AD achieves 93.70% accuracy, 91.51% recall, 92.19% precision, and 91.85% macro F1-score for three-class CN/MCI/AD classification. Furthermore, the framework provides interpretable regional trust scores that identify the anatomical regions contributing most strongly to each prediction.

Index Terms—Alzheimer’s disease, MRI, PET, multimodal learning, mixture of experts, trust-aware fusion, explainable AI

I. INTRODUCTION

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and one of the leading causes of dementia worldwide. Accurate diagnosis from neuroimaging data is essential for early intervention and clinical decision-making. Among available imaging modalities, MRI provides detailed structural information about brain atrophy, while PET captures complementary molecular abnormalities associated with disease progression [1], [2]. Their complementary nature has motivated extensive research in multimodal Alzheimer’s disease classification.

Recent deep learning approaches have demonstrated promising performance by combining multimodal neuroimaging information for AD diagnosis [5], [6]. However, most existing methods rely on global feature fusion, which may overlook region-specific pathological patterns and assume equal reliability across modalities and anatomical regions. Furthermore, different brain regions may contribute unequally to diagnosis, and their importance may vary across subjects and disease stages.

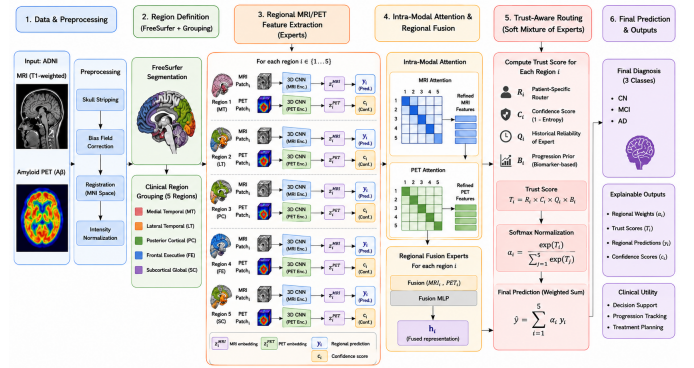


Fig. 1. Overview of TMREF-AD. MRI and PET volumes are partitioned into five clinically meaningful anatomical regions and encoded using modality-specific networks. Regional representations are refined through intra-modal attention, fused by regional experts, and aggregated using trust-aware expert fusion for Alzheimer’s disease classification.

To address these challenges, we propose **TMREF-AD**, a trust-aware multimodal regional expert fusion framework for Alzheimer’s disease classification. TMREF-AD combines region-aware multimodal representation learning, intra-modal attention, and trust-guided expert aggregation to improve both predictive performance and interpretability.

Contributions. This work: (i) introduces a regional expert framework for learning multimodal MRI-PET representations from five clinically meaningful brain regions; (ii) employs intra-modal attention to capture inter-regional dependencies; and (iii) proposes trust-aware fusion for adaptive aggregation and interpretable regional importance estimation.

II. METHOD OVERVIEW

Regional Partitioning. The proposed **TMREF-AD** framework performs multimodal Alzheimer’s disease classification using structural MRI and PET. First, MRI volumes are preprocessed and segmented using FastSurfer to obtain FreeSurfer-compatible anatomical brain labels [7]. These labels are grouped into five clinically meaningful macro-regions: medial temporal, lateral temporal, posterior cortical, frontal executive, and subcortical regions. The resulting regional masks are applied to both MRI and registered PET volumes to extract paired regional 3D patches.

Regional Feature Learning. For each region, modality-specific 3D CNN encoders independently extract MRI and PET representations. Intra-modal attention is then applied

separately within each modality to model dependencies among anatomical regions and refine regional embeddings [8]. Given regional features $\mathbf{h}_i$, the attention-refined representation is computed as

$$\tilde{\mathbf{h}}_i = \sum_{j=1}^N \alpha_{ij} \mathbf{h}_j, \quad \alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{k=1}^N \exp(s_{ik})}, \quad (1)$$

where s_{ij} denotes the similarity score between regions i and j , and N is the number of anatomical regions.

Regional Expert Fusion. The refined MRI and PET embeddings of each region are fused by a dedicated regional expert network inspired by mixture-of-experts learning [9]. For the i -th region, multimodal fusion is performed as

$$\mathbf{z}_i = \phi(W_m \tilde{\mathbf{m}}_i + W_p \tilde{\mathbf{p}}_i + b), \quad (2)$$

where $\tilde{\mathbf{m}}_i$ and $\tilde{\mathbf{p}}_i$ denote the attention-refined MRI and PET features, respectively, and $\phi(\cdot)$ represents a nonlinear activation function. Each expert produces a regional class prediction y_i together with a confidence estimate C_i .

Trust-Aware Expert Aggregation. To aggregate regional evidence, TMREF-AD employs a trust-aware expert fusion mechanism. The trust score of the i -th regional expert is defined as

$$T_i = R_i C_i Q_i, \quad (3)$$

where R_i denotes the patient-specific routing weight, C_i denotes prediction confidence, and Q_i denotes the historical reliability of the expert. The trust scores are normalized and used to compute the final prediction.

The framework predicts one of three diagnostic categories: cognitively normal (CN), mild cognitive impairment (MCI), or Alzheimer’s disease (AD). The learned trust scores further provide interpretable regional evidence by highlighting the anatomical regions that contribute most strongly to each diagnostic decision.

III. EXPERIMENTAL RESULTS AND FINDINGS

Experiments were conducted on subjects from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset [10]. The study cohort comprised 730 cognitively normal (CN), 570 mild cognitive impairment (MCI), and 300 Alzheimer’s disease (AD) subjects. Subject-level cross-validation was employed to prevent data leakage across multiple visits from the same participant.

For the CN/MCI/AD classification task, TMREF-AD achieved an accuracy of **93.70%**, a recall of **91.51%**, a precision of **92.19%**, a specificity of **94.88%**, and a macro F1-score of **91.85%**. As shown in Table I, the proposed framework outperforms representative Alzheimer’s disease classification methods, demonstrating the effectiveness of multimodal regional expert fusion.

Table II presents the ablation study. Removing intra-modal attention decreases the F1-score from 91.85% to 88.12%, highlighting the importance of modeling inter-regional dependencies within each modality. Replacing trust-aware expert

TABLE I
PERFORMANCE COMPARISON WITH REPRESENTATIVE ALZHEIMER’S DISEASE CLASSIFICATION METHODS.

Method	Modality	Accuracy	Recall	Precision	Specificity	F1
LDA-ELM [3]	MRI+PET+CSF+SNPs	66.70	—	—	—	64.90
Stacked DAE [4]	MRI+Demo+SNPs	78.00	78.00	77.00	—	78.00
PDMML [5]	MRI+PET+Demo	80.80	81.00	81.00	—	81.00
MCLNC [6]	MRI+Cognitive	90.44	86.29	88.97	93.47	87.47
TMREF-AD	MRI+PET	93.70	91.51	92.19	94.88	91.85

aggregation results in a larger performance degradation, reducing the F1-score to 83.59%. These findings confirm that both intra-modal attention and trust-aware fusion are essential components of TMREF-AD.

TABLE II
ABLATION STUDY FOR THE PROPOSED FRAMEWORK.

Variant	Accuracy	Recall	Precision	Specificity	F1
Full model	93.70	91.51	92.19	94.88	91.85
w/o intra-modal attention	91.08	88.34	87.91	90.72	88.12
w/o trust routing	85.31	83.42	83.76	84.63	83.59

Beyond predictive performance, the learned trust scores provide interpretable regional evidence by identifying the anatomical regions that contribute most strongly to each diagnostic decision.

IV. CONCLUSION AND FUTURE WORK

TMREF-AD combines regional mixture-of-experts learning, intra-modal attention, and trust-aware fusion for multimodal MRI-PET Alzheimer’s disease classification. Results on ADNI demonstrate strong performance and interpretable regional importance estimates. Future work will investigate longitudinal disease progression modeling, incorporation of additional biomarkers, and validation on larger multi-center cohorts.

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